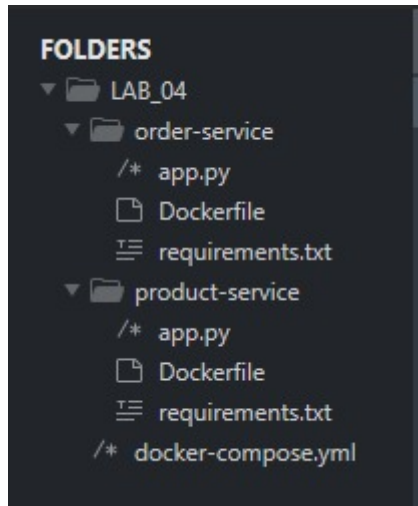


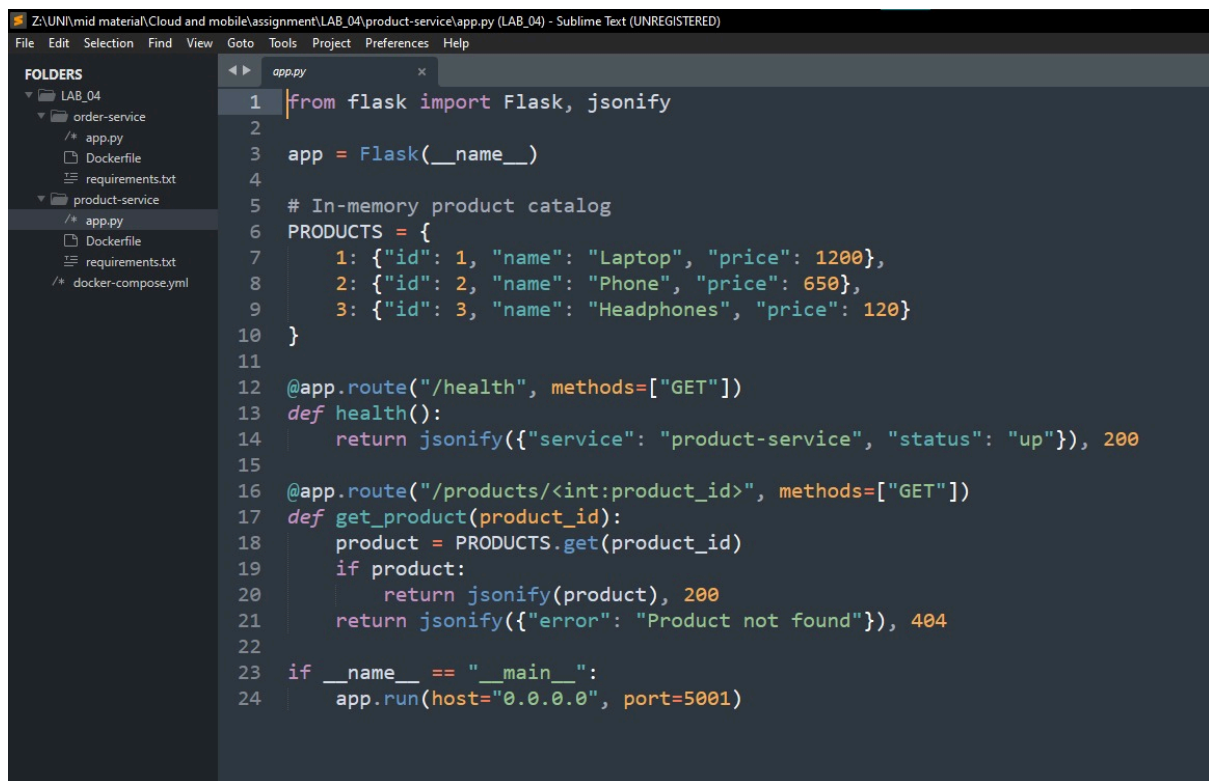
Ali El-Azouny(SE2)

LAB_04

PART A: (all folders and files are uploaded in the repository)



PART B: (Creating product-service/requirements.txt and product-service/app.py)



The screenshot shows the Sublime Text editor interface. The title bar indicates the file path: `Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04\product-service\requirements.txt`. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. On the left, the 'FOLDERS' sidebar shows a tree structure: `LAB_04` (expanded) contains `order-service` and `product-service`. `order-service` contains `/* app.py`, `Dockerfile`, and `requirements.txt`. `product-service` contains `/* app.py`, `Dockerfile`, and `requirements.txt`. The `requirements.txt` file under `product-service` is selected. The main editor area shows the content of this file: `1 flask==3.0.3`. The line number '1' is highlighted in the left margin.

PART C: (creating order-service/app.py and order-service/requirements.txt)

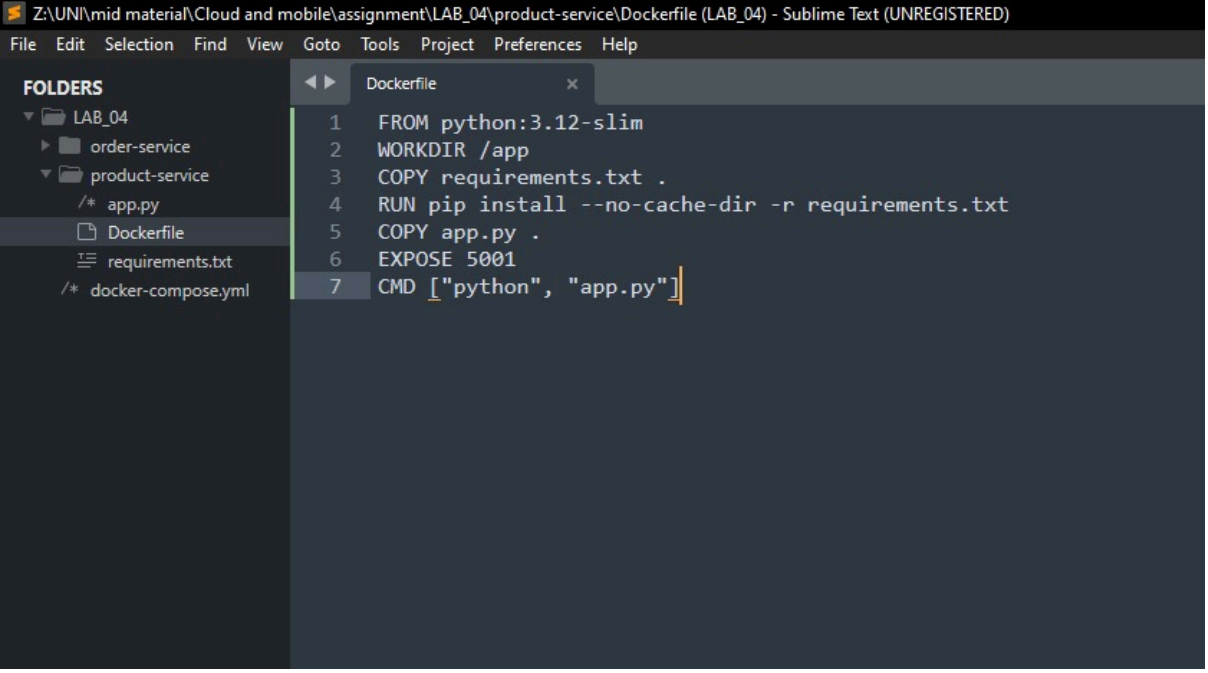
The screenshot shows the Sublime Text editor interface. The title bar indicates the file path: `Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04\order-service\requirements.txt (LAB_04) - Sublime Text (UNREGISTERED)`. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. On the left, the 'FOLDERS' sidebar shows a tree structure: `LAB_04` (expanded) contains `order-service` and `product-service`. `order-service` contains `/* app.py`, `Dockerfile`, and `requirements.txt`. `product-service` contains `/* docker-compose.yml`. The `requirements.txt` file under `order-service` is selected. The main editor area shows the content of this file: `1 flask==3.0.3` and `2 requests==2.32.3`. The line numbers '1' and '2' are highlighted in the left margin.

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04\order-service\app.py (LAB_04) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

FOLDERS
  LAB_04
    order-service
      /* app.py
      Dockerfile
      requirements.txt
    product-service
      /* docker-compose.yml

requirements.txt
app.py
1  import os
2  import time
3  import requests
4  from flask import Flask, jsonify, request
5
6  app = Flask(__name__)
7
8  # Configured via environment variable
9  PRODUCT_SERVICE_URL = os.getenv("PRODUCT_SERVICE_URL", "http://product-service:5001")
10
11 def fetch_product(product_id, retries=2, delay=1):
12     url = f"{PRODUCT_SERVICE_URL}/products/{product_id}"
13     for attempt in range(retries + 1):
14         try:
15             response = requests.get(url, timeout=2)
16             return response
17         except requests.exceptions.RequestException:
18             if attempt < retries:
19                 time.sleep(delay)
20             else:
21                 return None
22
23 @app.route("/health", methods=["GET"])
24 def health():
25     return jsonify({"service": "order-service", "status": "up"}), 200
26
27 @app.route("/orders", methods=["POST"])
28 def create_order():
29     data = request.get_json()
30     product_id = data.get("product_id")
31     quantity = data.get("quantity", 1)
32
33     response = fetch_product(product_id)
34
35     if response is None:
36         return jsonify({"error": "product-service unavailable"}), 503
37     if response.status_code != 200:
38         return jsonify({"error": "invalid product"}), 400
39
40     product = response.json()
41     total = product["price"] * quantity
42
43     return jsonify({
44         "message": "Order created",
45         "product": product["name"],
46         "quantity": quantity,
47         "total_price": total
48     }), 201
49
50 if __name__ == "__main__":
51     app.run(host="0.0.0.0", port=5002)
```

PART D: (creating product-service/Dockerfile and order-service/Dockerfile)

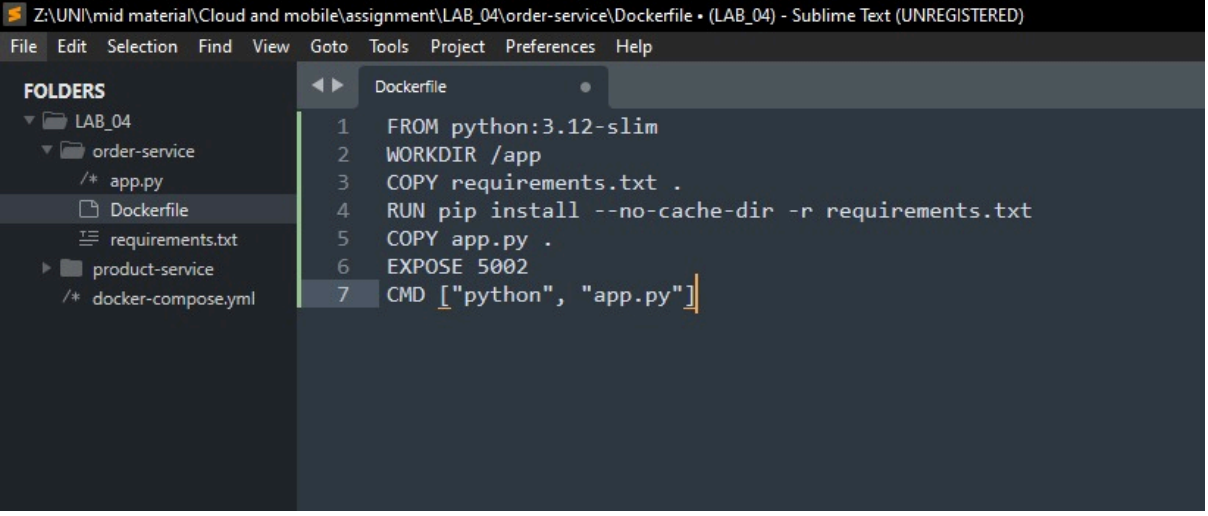


The screenshot shows the Sublime Text editor with the file path `Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04\product-service\Dockerfile (LAB_04) - Sublime Text (UNREGISTERED)`. The left sidebar displays the 'FOLDERS' panel with the following structure:

- LAB_04
 - order-service
 - product-service
 - app.py
 - Dockerfile (selected)
 - requirements.txt
 - docker-compose.yml

The main editor area shows the content of the selected `Dockerfile` with line numbers 1 through 7:

```
1 FROM python:3.12-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install --no-cache-dir -r requirements.txt
5 COPY app.py .
6 EXPOSE 5001
7 CMD ["python", "app.py"]
```



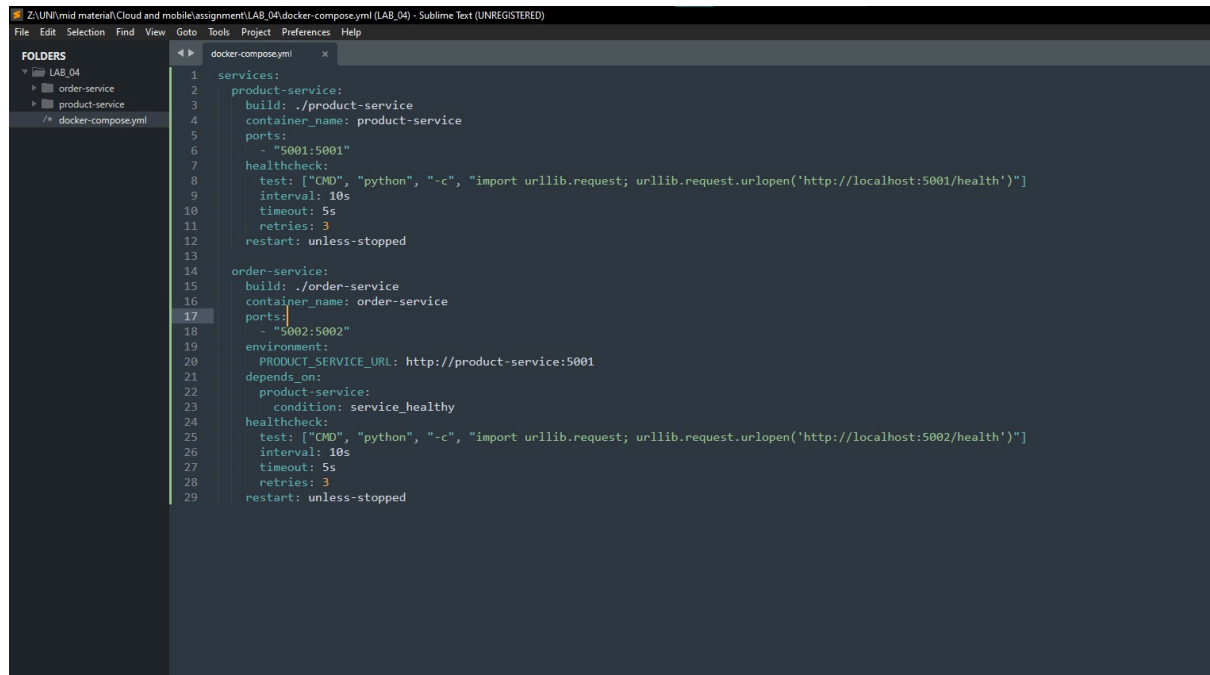
The screenshot shows the Sublime Text editor with the file path `Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04\order-service\Dockerfile • (LAB_04) - Sublime Text (UNREGISTERED)`. The left sidebar displays the 'FOLDERS' panel with the following structure:

- LAB_04
 - order-service
 - app.py
 - Dockerfile (selected)
 - requirements.txt
 - product-service
 - docker-compose.yml

The main editor area shows the content of the selected `Dockerfile` with line numbers 1 through 7:

```
1 FROM python:3.12-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install --no-cache-dir -r requirements.txt
5 COPY app.py .
6 EXPOSE 5002
7 CMD ["python", "app.py"]
```

PART E: (creating docker-compose.yml)



```
1  services:
2    product-service:
3      build: ./product-service
4      container_name: product-service
5      ports:
6        - "5001:5001"
7      healthcheck:
8        test: ["CMD", "python", "-c", "import urllib.request; urllib.request.urlopen('http://localhost:5001/health')"]
9        interval: 10s
10       timeout: 5s
11       retries: 3
12       restart: unless-stopped
13
14    order-service:
15      build: ./order-service
16      container_name: order-service
17      ports:
18        - "5002:5002"
19      environment:
20        PRODUCT_SERVICE_URL: http://product-service:5001
21      depends_on:
22        product-service:
23          condition: service_healthy
24      healthcheck:
25        test: ["CMD", "python", "-c", "import urllib.request; urllib.request.urlopen('http://localhost:5002/health')"]
26        interval: 10s
27        timeout: 5s
28        retries: 3
29        restart: unless-stopped
```


PART F: (building and running the system)

```
C:\Windows\System32\cmd.exe

=> [order-service internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 205B                                                            0.0s
=> [order-service internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                                  0.0s
=> [order-service internal] load build context                                                  0.1s
=> => transferring context: 1.64kB                                                              0.0s
=> [order-service 3/5] COPY requirements.txt .                                                  0.1s
=> [order-service 4/5] RUN pip install --no-cache-dir -r requirements.txt                    7.1s
=> [order-service 5/5] COPY app.py .                                                            0.1s
=> [order-service] exporting to image                                                            1.2s
=> => exporting layers                                                                           0.7s
=> => exporting manifest sha256:501457f96bb34bb1e3f7a6cfe43cc9d7c7d52208e2cee40bcb4c384569ea767a 0.1s
=> => exporting config sha256:8f7e7f9b84f605e692f23048b0a1cd153e18a614c3c2f3f92fd3d13993779054 0.0s
=> => exporting attestation manifest sha256:65f4c0d67628e98607a7c2b4e3982ff632cb6e12dd413293940ec88bf008af90 0.0s
=> => exporting manifest list sha256:6a0fe1197264a0fef125909bf8ac1575ffb7e1413b8d6036f67b96593409db89 0.0s
=> => naming to docker.io/library/lab_04-order-service:latest                                0.0s
=> => unpacking to docker.io/library/lab_04-order-service:latest                            0.3s
=> [order-service] resolving provenance for metadata file                                       0.0s

[+] Running 3/3
  Network lab_04_default          Created                                0.0s
  Container product-service      Healthy                          11.1s
  Container order-service        Started                       11.4s

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>
```

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker compose ps
NAME                IMAGE                COMMAND                SERVICE        CREATED        STATUS
PORTS
order-service       lab_04-order-service  "python app.py"        order-service   2 minutes ago  Up 2 minutes (healthy)
0.0.0.0:5002->5002/tcp
product-service     lab_04-product-service  "python app.py"        product-service 2 minutes ago  Up 2 minutes (healthy)
0.0.0.0:5001->5001/tcp
```

(checking logs)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker compose logs product-service
product-service | * Serving Flask app 'app'
product-service | * Debug mode: off
product-service | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
product-service | * Running on all addresses (0.0.0.0)
product-service | * Running on http://127.0.0.1:5001
product-service | * Running on http://172.19.0.2:5001
product-service | Press CTRL+C to quit
product-service | 127.0.0.1 - - [24/Apr/2026 21:39:31] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:39:41] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:39:52] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:02] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:12] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:22] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:32] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:42] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:40:52] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:02] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:12] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:22] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:32] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:43] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:41:53] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:03] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:13] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:23] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:33] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:43] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:42:53] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:43:03] "GET /health HTTP/1.1" 200 -
product-service | 127.0.0.1 - - [24/Apr/2026 21:43:13] "GET /health HTTP/1.1" 200 -
```

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker compose logs order-service
order-service | * Serving Flask app 'app'
order-service | * Debug mode: off
order-service | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
order-service | * Running on all addresses (0.0.0.0)
order-service | * Running on http://127.0.0.1:5002
order-service | * Running on http://172.19.0.3:5002
order-service | Press CTRL+C to quit
order-service | 127.0.0.1 - - [24/Apr/2026 21:39:42] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:39:52] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:03] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:13] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:23] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:33] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:43] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:40:53] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:03] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:13] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:23] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:33] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:43] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:41:53] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:03] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:14] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:24] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:34] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:44] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:42:54] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:43:04] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:43:14] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:43:24] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:43:34] "GET /health HTTP/1.1" 200 -
order-service | 127.0.0.1 - - [24/Apr/2026 21:43:44] "GET /health HTTP/1.1" 200 -
```

PART G: (functional testing)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>curl http://localhost:5001/health
{"service":"product-service","status":"up"}

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>curl http://localhost:5001/products/1
{"id":1,"name":"Laptop","price":1200}

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>curl http://localhost:5001/products/99
{"error":"Product not found"}
```

(test)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>curl -X POST http://localhost:5002/orders -H "Content-Type: application/json" -d '{"product_id": 1, "quantity": 2}'
{"message":"Order created","product":"Laptop","quantity":2,"total_price":2400}
```

PART H:(Failure injection and resilience observation)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker stop product-service
product-service

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>curl -X POST http://localhost:5002/orders -H "Content-Type: application/json" -d '{"product_id": 2, "quantity": 1}'
{"error":"product-service unavailable"}
```

(bringing the system to a healthy state again)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker start product-service
product-service

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_04>docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
order-service	lab_04-order-service	"python app.py"	order-service	11 minutes ago	Up 11 minutes (healthy)	0.0.0.0:5002->5002/tcp
product-service	lab_04-product-service	"python app.py"	product-service	11 minutes ago	Up 3 seconds (health: starting)	0.0.0.0:5001->5001/tcp